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Reviewer Name: Sankarnarayanan, Brahadeesh

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Address:

Department of Mathematics
Indian Institute of Technology Jodhpur
NH65, Nagaur Road, Karwar
Jodhpur Rajasthan, 342037
INDIA
briha1994@gmail.com

Author: Frankl, Peter; Wang, Jian

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Review text:

A family $\mathcal{F}$ of subsets of $[n]$ is said to be *p-intersecting* if $|F \cap F'| \geq p$ for all $F, F' \in \mathcal{F}$, and it is said to be *q-union* if $|F \cup F'| \leq n - q$ for all $F, F' \in \mathcal{F}$. The authors consider families $\mathcal{F}$ that are both *p-intersecting* and *q-union*, called *(p, q)-families*.

Katona [Combinatorics (Proc. NATO Advanced Study Inst., Breukelen, 1974), Part 2: Graph theory; foundations, partitions and combinatorial geometry, pp. 13–42, Math. Centre Tracts, No. 56, Mathematisch Centrum, Amsterdam, 1974; MR0351890] considered the extremal problem of determining $m(n, p, q)$, the maximum size of a (p, q) -family over $[n]$. It is known that $m(n, 1, 1) = 2^{n-2}$ by the work of Daykin–Lovász [J. London Math. Soc. (2) **12** (1975/76), no. 2, 225–230; MR0384559] and Seymour [Mathematika **20** (1973), 208–209; MR0347619], and $m(n, 1, q) = m(n, q, 1)$ was explicitly found by Frankl [J. Combinatorial Theory Ser. A **19** (1975), no. 2, 208–213; MR0376364] and Wang [J. Combinatorial Theory Ser. A **23** (1977), no. 3, 344–348; MR0457230]. Frankl [*Extremal set systems*, Ph.D. Thesis, Hungarian Academy of Science, 1977 (in Hungarian)] conjectured that the extremal (p, q) -families over $[n]$ are always attained as a product $\mathcal{G} \times \mathcal{H} := \{G \cup H : G \in \mathcal{G}, H \in \mathcal{H}\}$, where $\mathcal{G} \subset 2^X$ is *p-intersecting*, $\mathcal{H} \subset 2^Y$ is *q-union*, and $X \cup Y$ is a partition of $[n]$. The conjecture is easily verified when $n - p - q = 0$ or 1, and it was proved by Bang–Sharp–Winkler [Discrete Math. **45** (1983), no. 1, 123–126; MR0700856] when $n - p - q = 2$ or 3.

In this paper, the authors find the exact value of $m(n, p, q)$ for sufficiently

large n . Specifically, define the family

$$\mathcal{L}(n, a, b) := \{L \subset [n] : |L \cap [p + 2a]| \geq p + a, |L \cap [n - q - 2b + 1, n]| \leq b\}.$$

One can check that $\mathcal{L}(n, a, b)$ is of the form $\mathcal{G} \times \mathcal{H}$ for a p -intersecting family $\mathcal{G}$ and q -union family $\mathcal{H}$. The authors prove the conjecture when $n - p - q = 4$ by showing that if $\mathcal{F}$ is a (p, q) -family over $[n]$ where $n = p + q + 4$, then $|\mathcal{F}| \leq \max\{|\mathcal{L}(n, 2, 0)|, |\mathcal{L}(n, 1, 1)|, |\mathcal{L}(n, 0, 2)|\}$. They also prove that, when $n - p - q = 2r$, any (p, q) -family $\mathcal{F}$ over $[n]$ satisfies $|\mathcal{F}| \leq \max_{i+j=r} |\mathcal{L}(n, i, j)|$ for $n \geq 4r(r + 1)^2$. Similarly, they show that when $n - p - q = 2r + 1$, any (p, q) -family $\mathcal{F}$ over $[n]$ satisfies $|\mathcal{F}| \leq \max_{i+j=r} |\mathcal{L}(n, i, j)|$ for $n \geq 8r(r + 1)^2 + 6r$. In particular, this proves the conjecture for $n \geq (n - p - q + 1)^3$.

The authors use the technique of shifting due to Erdős–Ko–Rado [Quart. J. Math. Oxford Ser. (2) **12** (1961), 313–320; MR0140419], to restrict the analysis to initial families without loss of generality. For the case when $n - p - q = 2r$, they show that if a (p, q) -family $\mathcal{F}$ contains a *decisive set* G of type (a, b) , i.e., a set G for which $|[p] \setminus G| = a$ and $|[n - q + 1, n] \cap G| = b$, where $a + b = r$, then $\mathcal{F} \subset \mathcal{L}(n, a, b)$; otherwise, $|\mathcal{F}| < \max_{i+j=r} |\mathcal{L}(n, i, j)|$ for sufficiently large n . The case when $n - p - q = 2r + 1$ is a bit more subtle. If there are two decisive sets of different types in $\mathcal{F}$, then $\mathcal{F} \subset \mathcal{L}_0(n, a, b) := \{F \subset [n] : |F \cap [p + 2a]| \geq p + a, |F \cap [n - q - 2b - 1, n]| \leq b + 1\}$, which is not an optimal (p, q) -family. If there are two decisive sets G_1, G_2 of the same type (a, b) with $p + 2a + 1 \in G_1$ and $p + 2a + 1 \notin G_2$, then $\mathcal{F} \subset \mathcal{L}(n, a, b)$. Otherwise, $|\mathcal{F}| < \max_{i+j=r} |\mathcal{L}(n, i, j)|$ for sufficiently large n .